

# Sean Luka Girgis

Lead Data QA / Performance Testing Engineer | SQL, Data Pipelines & Production Validation

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## Professional Summary

Senior data quality, performance testing, and production validation professional with 20+ years of enterprise experience across SQL, data pipeline validation, performance testing, QA/testing workflows, production troubleshooting, log/metric analysis, Python automation, data quality checks, monitoring/APM, and stakeholder reporting. Strong background validating large operational datasets, investigating data discrepancies, comparing outputs to trusted baselines, identifying performance bottlenecks, documenting findings, and communicating risk clearly to technical and business teams. Best aligned to Lead Data QA / Data Tester roles requiring complex SQL, data pipeline validation, performance testing, production log/error investigation, QA leadership, ownership, and concise communication; Azure Portal/Application Insights/Containers, Snowflake, TypeScript, Playwright, C#, and formal code-quality metric platform ownership are positioned carefully as adjacent or learning areas rather than overstated production ownership.

## Professional Experience

### Senior Capacity & Data Engineer at CITI (2017-11 – 2025-12)

- Built and validated Python/Pandas/PySpark and SQL data pipelines ingesting telemetry from 6,000+ infrastructure endpoints into curated datasets for reporting, analytics, forecasting, and decision support.
- Developed data quality and regression-style checks to compare current outputs against prior trusted runs, identify abnormal metric movement, and prevent inaccurate data from reaching stakeholders.
- Used complex SQL queries across Oracle, Amazon Redshift, and reporting structures to validate transformations, investigate mismatches, reconcile outputs, and support root cause analysis.
- Investigated production pipeline issues by tracing source feeds, transformation logic, SQL behavior, metric behavior, failed outputs, missing or late data, latency symptoms, and downstream reporting discrepancies.
- Supported performance and reliability analysis across large-scale infrastructure using BMC TrueSight/TSCO, CA Wily/APM, AppDynamics, Oracle, AWS-backed reporting layers, and telemetry pipelines.
- Created KPI-based reporting and operational metrics that helped teams understand performance trends, bottleneck risks, utilization patterns, and service-health indicators.
- Documented validation logic, metric definitions, data flow notes, support procedures, runbooks, and troubleshooting steps to improve repeatability and team handoff.
- Partnered with infrastructure, application, analytics, engineering, and business teams to prioritize issues, explain findings, coordinate fixes, and improve operational reliability.

### Performance Engineer at G6 Hospitality LLC (2017-03 – 2017-11)

- Managed performance analysis and monitoring for Brand.com and critical hospitality systems using Dynatrace AppMon, Gomez Synthetic Monitoring, and HP Performance Center integration.

- Correlated load-test behavior with APM telemetry to identify bottlenecks, user-impacting symptoms, and improvement areas before production impact.
- Led the FAST project, mining real-user and synthetic monitoring metrics to identify performance improvement opportunities adopted by engineering teams.
- Created dashboards and reporting views that converted monitoring and test signals into actionable performance, reliability, and stakeholder insights.

#### **SME for CA APM (Senior Consultant) at CA Technologies / TIAA-CREF (2011 – 2016)**

- Served as CA APM/Introscope SME for a large financial-services environment with 50+ Enterprise Managers and 4,000–6,000 instrumented agents.
- Defined monitoring standards, dashboards, alert rules, SLA thresholds, validation checks, and operational reporting artifacts used by application, middleware, infrastructure, and operations teams.
- Built Perl and KornShell automation for extracting, validating, transforming, and distributing APM telemetry, replacing manual reporting workflows with repeatable quality checks.
- Provided mentoring, onboarding support, troubleshooting guidance, and documentation to improve team adoption of monitoring and performance-analysis practices.

#### **Performance Test Engineer at AT&T (2010-08 – 2011-07)**

- Developed and executed performance test plans for enterprise telecom applications, including baseline comparisons, regression review, NFR-style evaluation, and release-readiness analysis.
- Analyzed J2EE application behavior under load to identify throughput limits, JDBC bottlenecks, thread contention, heap pressure, CPU constraints, memory issues, and garbage-collection symptoms.
- Configured JMX Monitoring, Wily Introscope, and thread-dump automation scripts to improve runtime visibility during performance and regression testing.
- Documented test results, expected behavior, defects, risks, performance metrics, and remediation recommendations for engineering and release teams.

#### **Senior Systems & Data Migration Engineer at Sabre (2008-05 – 2010)**

- Led validation and performance-focused cutover support for a high-throughput shopping engine migration from 200+ MySQL nodes to a 6-node Oracle RAC cluster.
- Built C++/OC CI/OCI validation and regression testing utilities to verify correctness across millions of migrated transaction records.
- Optimized SQL and Oracle access patterns for high-volume workloads while reducing physical hardware footprint by 95%.
- Supported test planning, phased validation, troubleshooting, documentation, and production-readiness activities for a mission-critical enterprise migration.

#### **Architect / Developer (IRS CADE Project) at Computer Science Corporation (CSC) (2007-10 – 2008-05)**

- Designed UML class and sequence diagrams and developed modules for IRS CADE mainframe modernization.
- Built CICS, MQ Series, XML messaging interfaces, and DB2-backed components in a government modernization environment.
- Supported technical design documentation, integration patterns, test-readiness discussions, and structured data exchange across modernization components.

#### **Developer / Support Engineer (Billing, Interfaces, CSM/PRMS) at Corpus Inc. / Sprint (2001 – 2007)**

- Developed and supported billing, customer-management, and financial-style operational interfaces using Oracle, PL/SQL, C/C++, Pro\*C, SQL scripts, XML, and messaging patterns.
- Built automation scripts in KornShell/ksh, Perl, Awk, Syncsort, and SQL to support data extraction, transformation, reporting, deployments, validation, and production housekeeping.
- Provided production support, defect investigation, data maintenance, issue escalation, and troubleshooting for telecom billing and customer-management applications.
- Improved database performance by 10x through SQL, sequence, and process optimization while reducing memory usage by 75%.

## Education

### **Post-Graduate Diploma in Computer Engineering Technology / Computer Science**

Humber College, Toronto, Canada

### **Bachelor of Science in Civil Engineering**

Zagazig University, Egypt

## Skills

### Flagship Projects

#### **Production Data Quality and Regression Checks**

Built validation and comparison workflows for production reporting datasets, including row-count checks, null checks, missing feed diagnosis, unmatched join checks, and prior-run comparisons.

**Technologies:** Python, SQL, Pandas, Validation, Regression Checks, Runbooks

#### **FAST Performance Analytics Project**

Mined Dynatrace and Gomez Synthetic Monitoring data to identify optimization opportunities, improve transaction visibility, and communicate actionable performance findings to engineering teams.

**Technologies:** Dynatrace AppMon, Gomez Synthetic Monitoring, Performance Analytics, Dashboards, APM

#### **Enterprise APM Reporting Automation**

Built Perl and KornShell automation for extracting, validating, transforming, and distributing performance telemetry across large financial-services monitoring environments.

**Technologies:** CA APM, KornShell/ksh, Perl, SQL, Unix/Linux, Automation

#### **Serverless Lakehouse Platform (AWS)**

Designed an AWS data platform pattern using S3, Glue, Athena, Redshift, PySpark, and Parquet with validation checkpoints, SQL quality checks, and curated analytics datasets.

**Technologies:** AWS S3, AWS Glue, Athena, Redshift, PySpark, SQL, Data Quality